Maximal intersections in Suzuki torus normalizers

Human Compression project

Abstract

The subgroup definition of residual weak primitivity can be applied before assuming flag transitivity. For cyclic groups and odd dihedral groups it forces every parabolic to be maximal. In Suzuki groups, two successive maximal intersections then reduce the possible torus normalizers to compatible maximal subfields. Elementary size estimates exclude the remaining pairs. A separate normalizer calculation handles two subfields of the same size. These arguments prove the maximal-parabolic restrictions in Lemmas 4.2–4.6 and the full split-normalizer case of Lemma 5.1 in the fixed rank-three source.

1 The subgroup condition in cyclic and dihedral groups

For a family $(H_i)_{i\in I}$ in a finite group H, write $H_J = \bigcap_{j\in J} H_j$ and $H_\varnothing = H$. We use the condition

$$\text{for every } J \subsetneq I \text{ some } i \notin J \text{ makes } H_{J\cup\{i\}} \text{ maximal proper in } H_J. \tag{1}$$

The previously accepted strict-intersection lemma says that adding *any* further type strictly decreases an intersection. Its proof is short: $H_J \le H_i$ would imply $H_{I\setminus\{i\}} = H_I$, contrary to (1) at $I \setminus \{i\}$. In particular no distinct parabolics contain one another. The family induced inside any H_J again satisfies (1).

Lemma 1. *If H is cyclic and the family satisfies* (1)*, every H_i is maximal in H. Their indices are distinct primes.*

Proof. Induct on $|I|$. For one type this is the definition. Choose H_0 maximal, of prime index p. For $i \neq 0$, noncontainment in H_0 means that $|H_i|$ has the full p-part of $|H|$. Consequently $|H_i \cap H_0| = |H_i|/p$. The induced family in H_0 satisfies (1), so induction says that $H_i \cap H_0$ is maximal there. Its index is

$$[H_0 : H_i \cap H_0] = \frac{|H|/p}{|H_i|/p} = [H : H_i],$$

which is prime. Equal indices would give equal subgroups in a cyclic group, contrary to strict intersections. □

Lemma 2. *Let $H = D_{2m}$ with m odd. Every parabolic in a family satisfying* (1) *is maximal in H. At most one is cyclic. If the rank is at least three, at least two are dihedral.*

Proof. Again induct on the rank. Write $H = \langle r\rangle \rtimes \langle t\rangle$, with $|r| = m$, $t^2 = 1$, and $r^t = r^{-1}$. If a chosen maximal parabolic is $A = \langle r\rangle$, every other parabolic is dihedral by noncontainment. Its intersection with A is its rotation subgroup, which is maximal in A by Lemma 1 applied to the induced family. The parabolic therefore has prime index in H.

Otherwise choose $H_0 = \langle r^p, t\rangle$ for a prime $p \mid m$, after changing the reflection t. Every $H_0 \cap H_i$ is maximal in H_0 by induction (or Lemma 1 when $H_0 = C_2$). If $H_i = C_d$ is cyclic, noncontainment in H_0 gives the full p-part in d, and its intersection has order d/p. The only maximal cyclic subgroup of H_0 is its rotation subgroup, so $d/p = m/p$ and $H_i = A$.

If $H_i = \langle r^{m/d}, r^k t\rangle$ is dihedral, first suppose d has the full p-part of m. Then $p \nmid m/d$. The two reflection congruences have a simultaneous solution, so $H_0 \cap H_i$ is dihedral of order $2d/p$. Its index in H_0 is m/d; since it is maximal, that index is prime. Thus H_i is maximal in H. If d has a smaller p-part, then $d \mid m/p$. Noncontainment forces its reflection coset to be disjoint from H_0, so $H_0 \cap H_i = C_d$. Maximality in H_0 forces $d = m/p$, and again H_i is maximal, now of index p. This covers every subgroup. A maximal cyclic subgroup must be A, so there is at most one; the last assertion follows immediately. □

This proves directly the dihedral consequence cited from reference [20] in the source's Lemmas 4.3 and 4.4. It does not claim a new classification of all dihedral geometries or import their unexplained incidence properties.

2 Structural inputs and a subfield uniqueness lemma

Put $q = 2^n$, with $n \geq 3$ odd. We retain the classical Suzuki subgroup, ovoid and matrix description specified in the accepted L115 notes. Every proper subgroup either lies in a point stabilizer, a split torus normalizer, a nonsplit torus normalizer, or is itself a simple subfield group. Write

$$B_q = U_q \rtimes \mathbb{F}_q^*, \quad D_q = C_{q-1} \rtimes C_2, \quad N_\epsilon(q) = C_{a_\epsilon(q)} \rtimes C_4, \quad a_\epsilon(q) = q + \epsilon\sqrt{2q} + 1.$$

The three odd torus orders are pairwise coprime, and a nonidentity odd element lies in a unique torus. A proper simple subfield size satisfies $q \geq s^3$. The accepted elementary estimates imply, also directly,

$$q/2 \leq a_\epsilon(q) < 2q \quad (q \geq 8). \tag{2}$$

Indeed $q + 1 - \sqrt{2q} \geq q/2$ and $q + 1 + \sqrt{2q} < 2q$ in this range. The uniqueness of an embedded $D_{2(s-1)}$ or $C_{a_\delta(s)} \rtimes C_4$ inside a fixed-size subfield group was proved in the accepted normalizer-fibre note.

Lemma 3. *For $s = 2^d \geq 8$, $d \mid n$, every subgroup conjugate to the standard $Z_s \rtimes \mathbb{F}_s^*$ is contained in exactly one subgroup isomorphic to $\mathrm{Sz}(s)$. Here Z_s is the centre of a Sylow two-subgroup of the standard subfield group.*

Proof. We prove that the standard $H = Z_s \rtimes \mathbb{F}_s^*$ is self-normalizing in $G = \mathrm{Sz}(q)$. Its characteristic Sylow two-subgroup Z_s determines its unique fixed ovoid point. A normalizer therefore lies in B_q. Use the accepted coordinates $u(a, b)$ for its Sylow subgroup and $h(v)$ for the diagonal torus. Conjugation by $h(v)$ multiplies the central coordinate by $v^{\theta+2}$, where $\theta = 2^{(n+1)/2}$. Preserving the central line $\mathbb{F}_s$ forces $v^{\theta+2} \in \mathbb{F}_s^*$. Since $\theta+2$ is coprime to $q-1$, this is equivalent to $v \in \mathbb{F}_s^*$. Remove this diagonal factor.

For $u(a, b)$, preservation of H forces every commutator with $h(k)$, $k \in \mathbb{F}_s^*$, to have first coordinate zero. That coordinate is a nonzero scalar multiple of $(k^\theta - 1)a$. Taking $k \neq 1$ therefore gives $a = 0$. The remaining commutator coordinate is a nonzero scalar multiple of $(k^{\theta+2} - 1)b$ and must lie in $\mathbb{F}_s$. Again take $k \neq 1$; its coefficient is a nonzero element of $\mathbb{F}_s$, so $b \in \mathbb{F}_s$. Thus $\mathrm{N}_G(H) = H$.

The standard subfield group $S = \mathrm{Sz}(s)$ is self-normalizing, as proved in the accepted note. Its copies of the indicated affine subgroup form one conjugacy class: the Sylow centres are conjugate, and the odd complements in their point stabilizers are conjugate by the elementary affine-complement argument. The same incidence count used in that note therefore gives

$$\#\{S^g : H \leq S^g\} = \frac{|\mathrm{N}_G(H)|}{|\mathrm{N}_S(H)|} = 1.$$

Conjugation gives the assertion for every conjugate of H. □

We also need an elementary overgroup observation valid for every s. In $S = \mathrm{Sz}(s)$, the proper overgroups of its full split torus $A_s = C_{s-1}$ are exactly

$$A_s, \quad D_s, \quad Z_P A_s, \quad Z_Q A_s, \quad B_P, \quad B_Q, \tag{3}$$

where P, Q are the two fixed ovoid points. The classical list excludes nonsplit normalizers by coprimality and proper simple subfields by their smaller torus orders. Inside either point stabilizer an overgroup is VA_s. Full scalar multiplication acts irreducibly on both $Z(U)$ and $U/Z(U)$ over $\mathbb{F}_2$. Thus $V = 1, Z(U)$ or U: if its image on $U/Z(U)$ is nonzero, then $VZ(U) = U$, and the accepted identity $Z(U) = \Phi(U)$ gives $V = U$. This proves (3).

3 Excluding a full nonsplit normalizer

Theorem 4. *A family satisfying* (1) *in* $\mathrm{Sz}(q)$ *cannot have a full nonsplit torus normalizer as a parabolic when its rank is at least three.*

Proof. The prime-degree case is already accepted, so assume n composite. Both nonsplit torus orders are then composite by the accepted subfield arithmetic. Suppose $G_0 = N = C_m \rtimes C_4$, with $m = a_\epsilon(q)$. Choose $J = G_{01}$ maximal in N. Its possibilities are D_{2m} and $C_{m/p} \rtimes C_4$, $p \mid m$ prime. No proper subgroup outside N contains D_{2m}: the classical list and the torus partition exclude the other solvable types, and a proper simple subfield has every torus order less than $2s < q/2 \le m$. The first possibility is therefore excluded.

Set $d = m/p > 1$ and $J = C_d \rtimes C_4$. The proper overgroup G_1 is not contained in N by strict intersections. The classical list forces $G_1 = \mathrm{Sz}(s)$. Inside it the odd subgroup C_d lies in a nonsplit torus of order $a_\delta(s)$, which divides m and is smaller than m. Since it is a multiple of m/p, it equals d. Thus J is its full normalizer in G_1. Moreover G_1 is a maximal subfield group: an intermediate field would supply a strictly intermediate compatible torus order between d and m, impossible when $m/d = p$ is prime. Write $q = s^a$ with a an odd prime.

Choose another type 2 so that $K = G_{012}$ is maximal in J. If $K = D_{2d}$, then G_{02} contains it. A subgroup of N containing K has odd kernel of order d or m. In the first case it is contained in the unique $C_d \rtimes C_4$ containing K, namely J. In the second case it contains D_{2m}, which cannot be in the proper subgroup $G_2 \not\le N$. Both contradict strict intersections.

The other possibility is $K = C_{m/(pr)} \rtimes C_4$ for a prime $r \mid d$. Since G_{02} contains this fixed C_4, its odd kernel has order divisible by $m/(pr)$ and dividing m. Noncontainment in J requires the full p-part. If $r = p$, this forces the whole N, which is impossible. If $r \neq p$, it forces $G_{02} = C_{m/r} \rtimes C_4$. The same argument as for G_1 gives $G_2 = \mathrm{Sz}(t)$, with $m/r = a_\eta(t)$ and $q = t^b$, b prime.

If $s = t$, the two torus orders are distinct and hence are the opposite ones. Their product is $s^2 + 1$, whereas

$$m = \frac{(m/p)(m/r)}{m/(pr)} \le s^2 + 1.$$

This contradicts $m \ge q/2 \ge s^3/2 > s^2 + 1$. If $s \neq t$, the distinct maximal field indices a, b give $q = 2^{hab}$, $s = 2^{hb}$ and $t = 2^{ha}$ for a positive integer h. Distinct odd primes satisfy $ab - a - b \ge 7$, so $q/(st) \ge 128$. But p divides $m/r = a_\eta(t)$, while (2) gives

$$p = \frac{m}{a_\delta(s)} > \frac{q}{4s} \ge 32t > a_\eta(t).$$

This final contradiction proves the theorem. □

4 The full split normalizer

Theorem 5. *Suppose $q-1$ is composite and a family of rank at least three satisfying* (1) *has $G_0 = D_q$. At most one G_{0i} is dihedral.*

Proof. Apply Lemma 2 to the family induced in G_0. Every G_{0i} is maximal in D_q. Suppose two are dihedral: $G_{01} = D_{2m/p}$ and $G_{02} = D_{2m/r}$, where $m = q-1$ and p, r are primes. Their rotation subgroups are nontrivial. A proper overgroup of such a split dihedral subgroup cannot lie in a point stabilizer: its involution has trivial image in the abelian odd quotient, so cannot invert the nontrivial odd image of a rotation. Coprimality excludes nonsplit normalizers, and the torus partition identifies the only possible split normalizer as D_q. Consequently $G_1 = \mathrm{Sz}(s)$ and $G_2 = \mathrm{Sz}(t)$. Their split torus orders are multiples of m/p and m/r, respectively, and proper divisors of m. Hence

$$s - 1 = m/p, \qquad t - 1 = m/r.$$

As in the preceding proof, the prime indices imply that both are maximal subfield groups.

If $p \neq r$, then $s \neq t$. Write their distinct field indices as a, b. The same calculation gives $q/(st) \geq 128$. The prime p divides $m/r = t - 1$, but

$$p = \frac{q-1}{s-1} > \frac{q}{2s} \geq 64t > t - 1,$$

a contradiction.

If $p = r$, the two distinct dihedral subgroups have the same rotation subgroup $A_s = C_{s-1}$ and disjoint reflection cosets. Thus $G_{012} = A_s$, and $s = t$. Strictness requires $L = G_1 \cap G_2 > A_s$. Apply (3) inside G_1. If $L = D_s$, it equals the normalizer of A_s in G_1, namely G_{01}. Then $G_{012} = G_{01}$, a contradiction. Each other possibility contains one of the groups $Z_P A_s$ or $Z_Q A_s$. By Lemma 3, this subgroup is contained in a unique copy of $\mathrm{Sz}(s)$, so $G_1 = G_2$, again impossible. This treats the same-size case separately; distinct field indices were not assumed merely from distinct parabolics. □

Corollary 6. *A family satisfying* (1) *with a full split normalizer as a parabolic has rank at most three. In rank at least four, every parabolic maximal in* $\mathrm{Sz}(q)$ *must be a simple subfield group.*

Proof. If $q - 1$ is prime, the earlier accepted rank bound applies. If it is composite and the rank is at least four, the residue in D_q has rank at least three. Lemma 2 gives at least two dihedral parabolics in it, contrary to Theorem 5. For the last assertion, the accepted point-stabilizer exclusion and Theorem 4 exclude the other solvable maximal subgroups. The classical maximal-subgroup list leaves only simple subfields. □

Corollary 7. *Suppose n is composite and a rank-three family satisfying* (1) *has $G_0 = D_q$. After interchanging types 1 and 2, its pattern is*

G_0	G_1	G_2
$D_{2(q-1)}$	$E_q \rtimes C_{q-1}$	$\mathrm{Sz}(s)$

$$G_{01} = C_{q-1}, \quad G_{02} = D_{2(s-1)}, \quad G_{12} = E_s \rtimes C_{s-1}, \quad G_{012} = C_{s-1},$$

where $(q-1)/(s-1)$ is prime. The degree n is a power of one odd prime. This is a necessary pattern, not an existence assertion.

Proof. The two proper intersections in D_q are maximal by Lemma 2. At most one is dihedral by Theorem 5, and there cannot be two cyclic ones. Thus one is C_{q-1} and the other is $D_{2(q-1)/p}$. The proof of Theorem 5 gives $G_2 = \mathrm{Sz}(s)$ with $s - 1 = (q - 1)/p$. The full-torus overgroups (3), now in $\mathrm{Sz}(q)$, and strictness leave $G_1 = Z_P C_{q-1}$ or $Z_Q C_{q-1}$; the full point stabilizers have already been excluded. Their intersection with G_2 properly contains C_{s-1}. Applying (3) inside G_2, it cannot be dihedral, since a split dihedral group cannot lie in a point stabilizer. It cannot be the full B_s, whose Sylow two-subgroup has exponent four, whereas every two-subgroup of G_1 is elementary abelian. The remaining option is $E_s \rtimes C_{s-1}$.

For completeness, the prime quotient forces the degree assertion. Write $q = s^a$. If a were composite, the geometric sum $(s^a - 1)/(s - 1)$ would factor into two integers greater than one; thus a is prime. Write $s = 2^d$. If a prime $b \neq a$ divided d, put $x = 2^{d/b}$. The smaller integer $M = 1 + x + \cdots + x^{a-1} > 1$ divides $1 + x^b + \cdots + x^{b(a-1)}$: modulo M, $x^a = 1$, and multiplication by b permutes the residues modulo a. This would again make $(q - 1)/(s - 1)$ composite. Therefore d is a power of a, and so is $n = ad$. □

5 Attribution and scope

The source is D. Leemans, *The rank 3 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Note di Matematica 19(1) (1999), 43–63. Theorem 4 proves its Lemma 4.2 for every permitted field degree. Lemma 2, Theorem 5 and Corollary 6 prove Lemmas 4.3–4.6; Corollary 7 proves the precise necessary assertion in Lemma 5.1. The cyclic and dihedral induction replaces the explicitly cited consequence of the earlier dihedral classification. The same-size subfield argument is supplied explicitly rather than inferring distinct field degrees from distinct subgroups. These are elementary arguments; no priority claim is made for the methods.

The classical subgroup classification, the faithful Suzuki coordinates, ovoid action and subfield embeddings remain borrowed structural inputs. The accepted L115 notes supply the Frattini identity, scalar actions, torus arithmetic and normalizer-incidence counting. The accepted L114 prime-degree proof supplies strict intersections and the point-stabilizer and prime-degree exclusions. No finite computation or Lean certificate is asserted in this component. It does not repair the refuted complete candidate list in Theorem 5.5, prove the remaining thinness assertion, or confer whole-original or new-conjecture credit.